

◆ BLOCKCHAIN OSINT VISUALIZER

Follow the Money

ChainTrace traces Ethereum fund flows, detects suspicious patterns, enriches addresses with OSINT labels, and generates forensic investigation reports.

ACADEMIC PROTOTYPE

IIT DEMO

5-hop

MAX TRACE DEPTH

100+

RISK SIGNALS

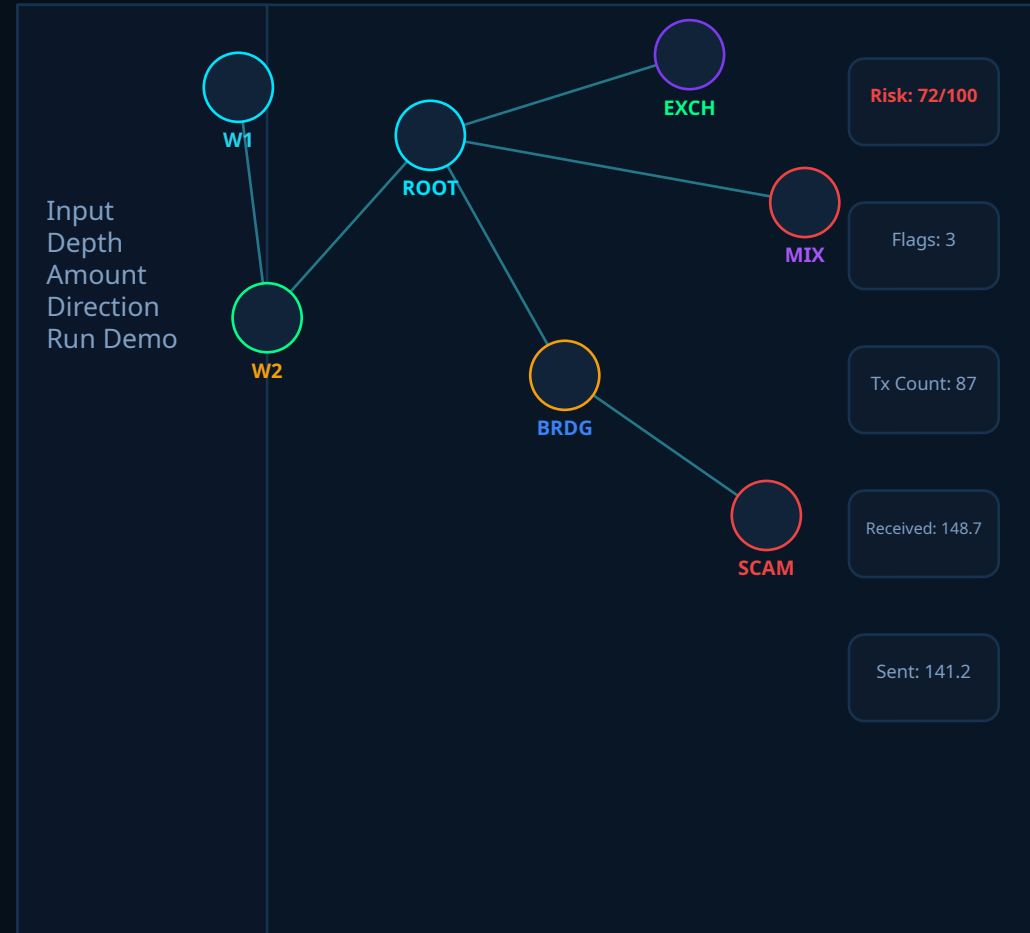
REAL-TIME

GRAPH ANALYSIS

OSINT

LABEL ENRICHMENT

INVESTIGATION DASHBOARD



PROBLEM

Why ChainTrace Matters

Blockchain investigations get messy fast: money moves across many wallets, hops through mixers or bridges, and becomes difficult to trace manually.



Pain Points

Manual tracing is slow. Multi-hop flows create complexity. Fund splitting hides destination patterns. Public labels are fragmented. Investigators need faster reporting.



Target Users

OSINT analysts, blockchain forensics teams, cybercrime investigators, compliance professionals, and academic security researchers.



Goal

Create a visual investigation tool that traces fund flow, scores suspicious behavior, enriches known addresses, and produces report-ready findings.

SOLUTION

Investigation-Grade Analysis

Built for analysts who need to follow illicit fund flows quickly and explain findings clearly during a demo.



Multi-hop Graph Tracing

Visualize wallet relationships up to 5 hops deep.



Real-time Risk Scoring

Rule-based engine detects suspicious behavior out of 100.



OSINT Label Enrichment

Cross-reference exchanges, mixers, scam wallets, and bridges.



Timeline Analytics

Spot behavioral timing clusters and volume spikes.



Investigation Report

Auto-generated report with findings and recommendations.



Zoom-level Graph

Reveal deeper wallet layers as you zoom in.

WORKFLOW

How It Works

The product flow on the website is simple, structured, and easy to follow in a hackathon submission.



IMPLEMENTATION

Tech Stack

The prototype is demo-friendly: it can work offline with mock data, while still showing a realistic architecture.

Frontend

React 18 + Vite + TailwindCSS

Graph

Cytoscape.js

Backend

Python FastAPI

Database

SQLite / JSON

Blockchain

Etherscan API (optional)

Reports

Auto-generated JSON

OFFLINE DEMO READY

LIVE API OPTIONAL

ANALYSIS LAYER

Risk Engine + Label Intelligence

ChainTrace combines a rule-based scoring engine with wallet labels so investigators can quickly prioritize suspicious activity.



Risk Signals

Rapid consecutive transfers (+18), high fan-out pattern (+16), mixer interaction (+20), known scam address (+12), and unusual volume spike (+6).



Label Categories

Exchange, mixer, scam or phishing, donation or foundation, bridge, and unknown. These labels add context to wallet behavior.



Output

Risk score out of 100, risk level, graph nodes and edges, OSINT-enriched context, and a report ID for structured investigation output.

COMPARISON

Why ChainTrace Is Better Than Etherscan

ChainTrace is more useful than Etherscan for investigation workflows because it combines tracing, intelligence, and reporting instead of only raw transaction lookup.

- Multi-hop tracing up to 5 hops

- Risk scoring for suspicious behavior

- OSINT label enrichment

- Auto-generated investigation reports

- Zoom-level graph exploration for complex wallet networks